

Team 16: Online Internship and Placement Prep Portal
Requirement List (shared contract for SRS, SAD and Test Plan)

User classes: Student, Recruiter, Admin. IDs use the prefix IPP and must be used exactly as written in the SRS, the architecture traceability and the test case mapping.

Common (accounts and access control)

ID	Requirement
IPP-F-050	The system shall allow a user to register as a Student or a Recruiter by providing name, email address and password.
IPP-F-051	The system shall validate the email format at registration and reject an email address that is already registered, displaying an error message.
IPP-F-052	The system shall authenticate a registered user by email and password and start a session on successful login.
IPP-F-053	The system shall restrict each function to the roles allowed to use it (Student, Recruiter, Admin).
IPP-F-054	The system shall allow an Admin to approve a Recruiter account before that Recruiter can post internship listings.

Internship listings

ID	Requirement
IPP-F-001	The system shall allow a Recruiter to create an internship listing with title, company, description, location, stipend, duration, application deadline and required skills.
IPP-F-002	The system shall allow a Recruiter to edit or close only the listings the Recruiter created.
IPP-F-003	The system shall display active internship listings to Students in a paginated list.
IPP-F-004	The system shall allow a Student to search listings by keyword matched against title, company and required skills.
IPP-F-005	The system shall allow a Student to filter listings by location, duration, stipend range and required skills.
IPP-F-006	The system shall not show a listing as active after its application deadline has passed.
IPP-F-007	The system shall allow a Student to open a listing to view its full details.
IPP-F-008	The system shall allow an Admin to remove a listing that violates portal guidelines.

Applications

ID	Requirement
IPP-F-010	The system shall allow a Student to apply to an active listing by uploading a resume and an optional cover note.
IPP-F-011	The system shall reject a second application by the same Student to the same listing and display a message stating that the Student has already applied.

IPP-F-012	The system shall reject an application submitted after the listing deadline.
IPP-F-013	The system shall accept only PDF resumes up to a maximum file size of 2 MB and reject other files with an error message.
IPP-F-014	The system shall allow a Student to withdraw an application before a decision has been made on it.
IPP-F-015	The system shall allow a Recruiter to view the list of applicants for the listings the Recruiter created.
IPP-F-016	The system shall allow a Recruiter to set an application status to Under Review, Shortlisted, Rejected or Selected.
IPP-F-017	The system shall notify a Student within the portal when the status of the Student's application changes.

Progress tracking

ID	Requirement
IPP-F-020	The system shall display to a Student a dashboard listing all of the Student's applications with their current status.
IPP-F-021	The system shall display the status history of each application as a timeline with the date of each change.
IPP-F-022	The system shall display to a Student the percentage of available aptitude tests attempted in each category.

Aptitude tests

ID	Requirement
IPP-F-030	The system shall allow an Admin to create an aptitude test with a category (quantitative, logical or verbal), a time limit, and questions each with answer options and one correct answer.
IPP-F-031	The system shall allow a Student to browse available aptitude tests by category.
IPP-F-032	The system shall display a countdown timer while a Student attempts a test.
IPP-F-033	The system shall automatically submit a test attempt when its time limit expires.
IPP-F-034	The system shall automatically score a submitted attempt and store the score against the Student and the test.
IPP-F-035	The system shall display the result of an attempt to the Student after submission, showing for each question whether the answer was correct and what the correct answer was.
IPP-F-036	The system shall allow a Student to attempt a test more than once and shall store each attempt separately.

Analytics

ID	Requirement
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IPP-F-040	The system shall display to a Student the trend of the Student's scores across attempts for each test category.
IPP-F-041	The system shall identify the test category in which a Student has the lowest average score and display it as a weak area.
IPP-F-042	The system shall display to a Recruiter, for each of the Recruiter's listings, the number of applications received and the distribution of application statuses.
IPP-F-043	The system shall display to an Admin the total number of registered users, active listings, applications and test attempts.

Non-functional requirements (target values are proposed and can be adjusted by the team)

ID	Requirement
IPP-NF-001	Performance: the system shall return listing search results within 3 seconds for at least 95% of requests with 100 concurrent users.
IPP-NF-002	Performance: the system shall score a submitted test attempt and display the result within 2 seconds.
IPP-NF-003	Availability: the system shall be available at least 99.5% of the time in a month, excluding planned maintenance.
IPP-NF-004	Capacity: the system shall support at least 1,000 registered users and 200 concurrent users while still meeting IPP-NF-001.
IPP-NF-005	Usability: in a usability test, at least 90% of first-time Students shall be able to find and apply to an internship within 5 minutes without assistance.
IPP-NF-006	Compatibility: the system shall work on the latest two versions of Chrome, Firefox and Edge, and the interface shall be usable at screen widths of 360 px and above.
IPP-NF-007	Reliability: while a Student is attempting a test, the system shall save the Student's answers at least every 30 seconds so that at most 30 seconds of answers are lost if the connection drops.
IPP-NF-008	Maintainability: automated unit tests shall cover at least 70% of the backend code.

Security objectives

ID	Requirement
IPP-SO-1	Confidentiality: personal data and resumes of Students shall be accessible only to the owning Student, the Recruiter of the listing applied to, and Admins.
IPP-SO-2	Integrity: test scores and application statuses shall not be modifiable by Students or by anyone other than the authorised system function or role.
IPP-SO-3	Access control: only authenticated users shall access the portal, and each user shall be able to perform only the functions permitted to their role.

Security requirements

ID	Requirement
IPP-SEC-001	The system shall store passwords only as salted hashes and never in plain text.

IPP-SEC-002	The system shall transmit all data between client and server over HTTPS using TLS 1.2 or higher.
IPP-SEC-003	The system shall expire a user session after 60 minutes of inactivity.
IPP-SEC-004	The system shall verify the user's role on the server for every request and return an authorisation error when the role is not permitted.
IPP-SEC-005	The system shall validate and sanitise all user inputs to prevent injection and cross-site scripting attacks.
IPP-SEC-006	The system shall allow a resume file to be downloaded only by the owning Student, the Recruiter of the listing applied to, and Admins.
IPP-SEC-007	The system shall temporarily lock an account for 15 minutes after 5 consecutive failed login attempts.
IPP-SEC-008	The system shall compute test scores on the server and shall not send correct answers to the client before the attempt is submitted.